

Capstone-Project-Career-Preparation

Project Description:-

SmartExpense is a comprehensive, production-ready full-stack web application developed using Python and Flask. The project is designed to help users manage their personal finances efficiently by allowing them to track income, expenses, budgets, and financial goals in one centralized system. The application provides a secure user authentication system where users can register, log in, and manage their personal financial information. Users can add income and expense transactions, organize them into different categories, and monitor their overall financial balance. The SmartExpense dashboard provides a clear overview of the user's financial situation by displaying total income, total expenses, available balance, savings rate, and recent transactions. Interactive charts and visual reports help users understand their spending patterns and make better financial decisions.

What I Learned:-

- I learned how to build a complete full-stack web application using Python and Flask.
- I improved my understanding of fundamental Python concepts such as variables, data types, conditions, loops, functions, and modules.
- I learned how to apply Object-Oriented Programming (OOP) concepts using classes and objects.
- I learned how to design and manage databases using SQLAlchemy ORM.
- I gained experience in creating database models for users, transactions, budgets, and financial goals.
- I learned how to implement secure user registration, login, logout, and authentication systems using Flask-Login.

Features of the Program:-

1. Home Page
2. Delete Projects
3. About Me Section
4. Admin Login
5. Skills Section
6. Logout Functionality
7. Project Showcase
8. Admin Dashboard
9. Add Projects
10. SQLite Database
11. Edit Projects
12. Dynamic Content

OUTPUT:-

SMARTEXPENSE DASHBOARD

Welcome, User!

Total Income: ₹75,000.00

Total Expenses: ₹32,450.00

Available Balance: ₹42,550.00

Output:

EXPENSES BY CATEGORY

Food & Dining	₹10,000
Transport	₹8,000
Shopping	₹6,000
Bills & Utilities	₹4,000
Entertainment	₹2,500
Others	₹1,950

RECENT TRANSACTIONS

Salary	+₹75,000
Grocery Shopping	-₹2,500
Restaurant	-₹800
Freelance Work	+₹10,000
Electricity Bill	-₹1,200

Conclusion:-

This project helped me combine the Python concepts learned throughout the course and apply them to a real-world problem. It improved my understanding of Python programming, web development, databases, APIs, security, testing, and deployment, while also giving me experience in building a professional and portfolio-ready full-stack application.

The project uses SQLAlchemy for database management and stores information such as users, transactions, budgets, and financial goals. Flask-Login is used to implement secure authentication and protect user-specific data. The application also includes REST API functionality, allowing financial data to be accessed programmatically.